

# KNOWLEDGE KNOW-HOW

## Tech Stack Overview & Practical Implementation Report

**Name:** Favas M

**Role:** Full-Stack Developer / Cyber Security / AI / Blockchain

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### Professional Summary

I work with a practical mix of frontend frameworks, backend technologies, databases, AI tools, and cloud platforms to build modern web applications. My focus is on clean architecture, responsive UI, scalable backend logic, and efficient AI-assisted development. The technologies below are ones I use regularly in real projects and daily workflow.

## 1. Development Environment & Coding Tools

### Tools

- **VS Code** — Primary code editor for development, debugging, and project management
- **Cursor** — AI-assisted coding, faster iteration, and code support
- **Antigravity** — AI-powered workflow support and experimentation
- **Git & GitHub** — Version control, collaboration, and code tracking
- **Node.js & npm** — Runtime and package management
- **Postman** — API testing and endpoint debugging
- **Chrome DevTools** — UI inspection, debugging, and performance checks
- **Terminal / CLI** — Project setup, dependency installation, and Git commands
- **Notion** — Documentation, planning, and task tracking
- **Figma** — UI design, wireframes, and prototypes
- **Google Stitch** — Design inspiration for modern, polished interfaces

### How I use them

- Plan project structure in **Notion**
- Design UI flow in **Figma**
- Write and debug code in **VS Code**, **Cursor**, and **Antigravity**
- Test APIs in **Postman**
- Inspect frontend issues in **Chrome DevTools**

- Manage code history and collaboration through **GitHub**

## 2. Web Frameworks

### Frontend

- **React.js** — Component-based UI development
- **Next.js** — SSR, routing, performance, and scalable app structure
- **Tailwind CSS** — Fast, responsive, and consistent styling

### Backend

- **Node.js (Express.js)** — REST API development and backend logic
- **Python Django** — Structured backend development and data handling
- **Firebase** — Authentication, hosting, and lightweight backend support

### How I use them

- Build reusable interfaces using **React**
- Create production-ready apps using **Next.js**
- Style responsive layouts with **Tailwind CSS**
- Use **Django** for structured backend systems
- Use **Express.js** for lightweight APIs
- Use **Firebase** when real-time support or quick backend integration is needed

## 3. AI / ML Frameworks

### Tools

- **TensorFlow** — Basic understanding of ML concepts and workflows
- **Scikit-learn** — Basic data analysis and beginner ML use
- **OpenAI APIs** — Practical AI integration in applications

### How I use them

- Understand basic machine learning concepts
- Explore how AI features can be integrated into real applications
- Use API-based AI features in chatbot-style and assistant-like systems
- Support AI-driven interfaces and automation workflows

## 4. Agentic AI Tools ⚡

### Tools

- **LangChain** — Basic AI workflow construction
- **AutoGPT / Agent frameworks** — Conceptual understanding of agent-based systems
- **Custom prompt workflows** — Automation and task execution support

### How I use them

- Break large tasks into smaller steps
- Build structured prompt-based workflows
- Improve productivity using AI-assisted task execution
- Explore intelligent application behavior and automation

## 5. LLM Platforms 🧠

### Platforms

- **OpenAI (ChatGPT, GPT APIs)** — Coding help, debugging, planning, and AI integration
- **Google AI Studio (Gemini)** — Prompt-based app building and prototyping
- **Claude (Anthropic)** — Structured outputs, reasoning, and code generation

### How I use them

- Generate and refine code ideas
- Debug and improve logic
- Test prompt behavior
- Support project planning and architecture decisions
- Compare outputs before final implementation

## 6. AI Tools Frequently Used 🛠️

### Tools

- **ChatGPT** — Coding, debugging, learning, and idea generation
- **Claude** — Structured code generation and refinement
- **GitHub Copilot** — Inline coding assistance
- **Google AI Studio** — AI prototyping and feature testing
- **Notion AI** — Documentation and planning support

### How I use them

- Speed up coding and problem-solving

- Refine logic and implementation steps
- Improve project planning and note organization
- Prototype AI-driven features faster
- Support daily development and debugging tasks

## 7. Cloud Platforms

### Platforms

- **Vercel** — Frontend deployment for Next.js and React applications
- **Netlify** — Static site hosting
- **Firebase Hosting** — Web app deployment and backend support
- **Render** — Basic backend deployment

### How I use them

- Deploy projects quickly and reliably
- Host frontend and full-stack applications
- Manage production builds and environment settings
- Publish live project versions for testing and submission

## 8. Databases

### Databases

- **MongoDB** — Primary NoSQL database
- **Firebase Firestore** — Real-time database
- **MySQL** — Relational database fundamentals
- **PostgreSQL** — Structured relational data and scalable systems

### How I use them

- Store and manage structured and unstructured data
- Choose databases based on project needs
- Use Firestore for real-time applications
- Use MySQL and PostgreSQL for management systems and relational workflows
- Use MongoDB for flexible content-based applications

## 9. Project Experience & Implementation

### Cambuzz – College Management System

**Stack:** React, Tailwind CSS, MySQL, Python Django

**Usage:** Built a structured management system for academic workflows. React and Tailwind CSS were used for the interface, Django for backend logic, and MySQL for student and

institutional data.

**What it demonstrates:** Full-stack system building, dashboard development, and database-driven application design.

## NextBridge Learning

**Stack:** Next.js, Python Django, PostgreSQL

**Usage:** Built an education-focused platform with optimized routing and structured backend logic. Next.js handled the frontend, Django managed server-side processing, and PostgreSQL supported relational data.

**What it demonstrates:** Performance-focused web development and strong backend structure.

## Modern Blog Application

**Stack:** Next.js, Mock API

**Usage:** Created a content-based blog platform using Next.js for rendering and routing, with Mock API for testing content flow.

**What it demonstrates:** SSR-style application design, API handling, and editorial UI presentation.

## AI Therapist Web App

**Stack:** React, Firebase

**Usage:** Built an AI-style chat application with React for the interface and Firebase for backend support and data handling.

**What it demonstrates:** Real-time interaction, AI-inspired user flow, and practical frontend integration.

## Movie Search Application

**Stack:** React, Firebase, OMDb API

**Usage:** Developed a movie discovery app using React for the UI, OMDb API for live movie data, and Firebase for support and hosting.

**What it demonstrates:** API integration, search logic, and dynamic UI rendering.

## News Listing Application

**Stack:** React, Firebase, Mediastack API

**Usage:** Built a news listing application using React for the frontend, Mediastack API for news data, and Firebase for backend support.

**What it demonstrates:** Live content fetching, listing-based UI design, and responsive application structure.

## 10. Everyday Workflow

My daily workflow combines these tools in a practical sequence:

**Notion** → Planning and documentation

**Figma** → UI design and layout

**VS Code / Cursor / Antigravity** → Coding and debugging

**React / Next.js / Django** → Development

**ChatGPT / Claude / Google AI Studio / Copilot** → AI support

**MongoDB / MySQL / PostgreSQL / Firestore** → Data management

**Vercel / Firebase** → Deployment

This workflow helps me stay organized, move faster, and build cleaner applications with better structure and output quality.

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## Conclusion

I regularly apply these technologies in real projects and everyday development work. My focus is on building scalable, user-friendly, and production-ready applications while continuously improving my skills across full-stack development, AI tools, cloud deployment, and database handling.